# Design & Reflection Parameters of Ultra-Wideband (UWB) Vivaldi Antenna

IEEE AESS Senior Capstone Design Project Proposal

## 1 Antenna Geometry & Parametric Architecture

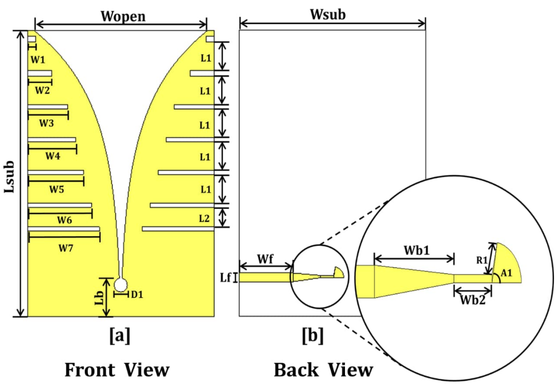
An Ultra-Wideband (UWB) Slotted Tapered Slot Vivaldi Antenna was designed for radar imaging and wideband sensing applications [1]. The structure features an exponentially tapered flare with side slot stubs ($W_1 - W_7$) to enhance low-frequency radiation performance, alongside a microstrip-to-slotline transition feed structure featuring radial and stub matching terminations ($R_1, A_1, W_{b1}, W_{b2}$).

Table 1: Geometric Design Parameters for Front and Back View Features

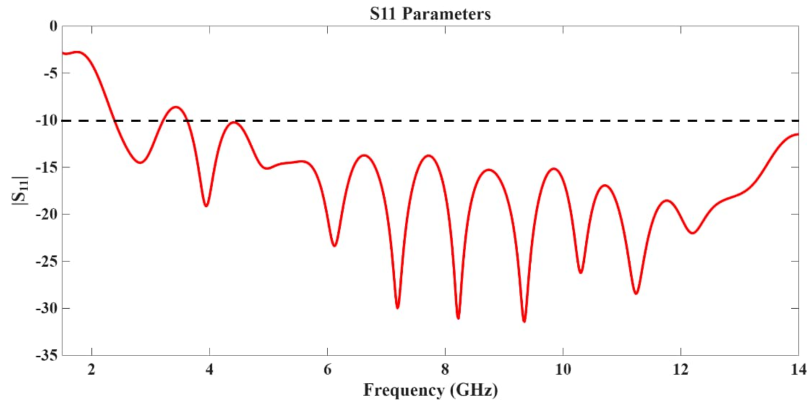
| Design Parameter | Symbol / Description | Feed & Transition Parameter | Symbol / Description |
|---|---|---|---|
| Substrate Length / Width | $L_{\text{sub}}$, $W_{\text{sub}}$ | Microstrip Feed Width / Length | $W_f$, $L_f$ |
| Aperture Opening Width | $W_{\text{open}}$ | Tapered Microstrip Matching | $W_{b1}$, $W_{b2}$ |
| Exponential Slot Cavity | $D_1$, $L_b$ | Radial Stub Radius / Angle | $R_1$, $A_1$ |
| Side Slot Widths | $W_1, W_2, W_3, W_4, W_5, W_6, W_7$ | Slot Spacings / Widths | $L_1$, $L_2$ |

## 2 Vivaldi Antenna Schematic & Reflection Coefficient ($S_{11}$)

The physical structure and corresponding return loss response ($|S_{11}|$) were evaluated across 2.5 GHz – 14.0 GHz to verify wideband operation below the −10 dB impedance matching threshold [1].




**(a) Antenna Geometry: Front View [a] with Corrugations and Back View [b] with Microstrip Feed Details**




**(b) Simulated Reflection Coefficient $|S_{11}|$ Across 2.5 to 14.0 GHz (−10 dB Limit Line)**

Figure 1: High-resolution UWB Vivaldi antenna schematic geometry and simulated reflection coefficient response.

## 3 Performance Summary & Impedance Bandwidth

The reflection coefficient curve demonstrates ultra-wideband operational characteristics:

$$\text{Impedance Bandwidth: } |S_{11}| \leq -10 \text{ dB} \quad \text{from } 2.5 \text{ GHz to } > 14.0 \text{ GHz} \tag{1}$$

Table 2: Resonance Points and Reflection Loss Performance Summary

| Frequency Band | Lowest Resonance | Mid-Band Minima | High-Band Minima | Upper Limit | Operating Bandwidth |
|---|---|---|---|---|---|
| $\|S_{11}\|$ **Value** | −19.1 dB (3.9 GHz) | −31.0 dB (8.2 GHz) | −31.5 dB (9.4 GHz) | −12.0 dB (14.0 GHz) | 2.5 GHz – 14.0 GHz (> 11.5 GHz) |

The corrugated side slots ($W_1 - W_7$) successfully suppress cross-polarization and extend the lower edge of the passband to 2.5 GHz, yielding superior return loss (< −15 dB) across most of the target band.

## References

[1] https://drive.google.com/file/d/1y79u657jSn$_{Y}SfsvMuCy1hEFyXT31fMp/view?usp = sharing$